Multenions and Differential Invariants.- II.

By Alex. McA ulay, Professor of Mathematics in the University of Tasmania.

(Communicated by W. B. Hardy, Sec.R.S. Received June 22, 1921.)

'§ 11. Thefundamental covariant vector Unity, The meaning of a manifo as based on a quadratic differential form is here taken for granted, and this,

whether, in the ordinary terminology, all or only some of its dimensions are real. Let

ia = ia, {a -1, 2, ... c)
ii, - ib\f (-1), (b - -f-1, -j- 2, ... nij ! ^^

n
p= X rhth
h= 1 J

e having any given value from 1 to n. In the present paper we shall deal with nmltenions and multenion linities based on a system of vectors p for which all the co-ordinates Xh and the similar scalars are real. In the case of relativity n -4, c= 3.

The fundamental quadratic form is taken to be

ds2 -Yodpydp = Vo

the sign which is given to ds2, a scalar, being convenient in applications to relativity, but inconvenient in applications to pure geometry, y, a self conjugate vector Unity, is a given function of position, a vector, an arbitrary function of p ; and since the transformation from y to y is given by (2) the fundamental Unity y is said to be covariant, [y and ds2 are real, but sometimes ds and sometimes dsff(-is real time-like interval " ; when d sff(-1) is real it is a length of reality is left undecided ds is an interval.] At any given position may be so chosen that y'dp'- dp',that is singularities; it has n mutually orthogonal principal directions and all its roots are real and positive, excluding zero values. Thus yi is assumed to have the same orthogonal system and its roots are taken to be the positive values of the square roots of y so that y%is real and without ambiguity.

We may, and from here on shall, take our previously introduced scalar k (§ 8) to be | y* \the determinant of yi. [We have

y = x f X

so that | VI- l ^ l x l%l8 = |VI x(^/A;')2,

because k/fe was defined by

kdb - k'dV

or |%|.]

It will be seen that when, in the usual notation of relativity, it is said that with Galilean co-ordinates

(^11, *722> {/23> "h^)>

we have not that 17(41, t%, 43, u)= (-4i>-4g,-*3,+ 4) but instea statement that 17 = 17* = 1 or ■

77(tl, 42, 43, i4) = (4l, 42, 43, 44),

for it will be observed that this gives

ds*- -d t f -dx£-dx32+ dscf.

A little care is required in expressing a linity, its conjugate, reciprocal, etc., by scalars or co-ordinates. It seems sufficient to illustrate from vector Unities. When we say that (f> is given by the square array of 16 scalars

a	b	c	
e	f	9	h
l	m	n	V
2	r	s	t

we imply that the co-ordinates of <f>ii are found in the first column thus

<^)4i = ai\+ £42*1"443+ (^

and so for the other columns. In the square array of scalars specifying (f>,

n

let city be the scalar in the xth row and yth column. Thus = axyix, or

1

more conveniently

W~xy - 0 i-x^

Except when c = nt

he square array of a self-conjugate linity i symmetrical about the principal diagonal. Let a^, , be the square arrays of </> and its conjugate </>'. By the meaning of conjugate we have

V04^4y - 4a;.

Thus instead of ayx= a^(as we have when

uyx - ix(4)

and in particular, when cf> is self-conjugate ayx -

If we would retain the 16 scalars gab with their usual meanings as in treatises on relativity we may use g(ai) for the corresponding scalars specifying 17. Thus from the fact that ds2 = and from (3)

9abVo4a77«4, - ^0

wheUCe dab = (5)

More details with regard to these notations and the theory of tensors will be found below in § 17.

' Noteon Notations in the Present Paper.-There will be many exceptions below to the conventions here to be described. Especially do exceptions occur in very frequently recurring symbols. Thus none of the following receive the " cross " notation although they should do according to the conventions,

\ Ea, Ea. Cw, C,

A. is a covariant vector, and of the other four C*,, C are contramixt linities while E,,, Ea are portions of contramixt linities. The explanations of the technical terms will be made as they come to be used below.

(1) Contra variant multenions will have no distinguishing mark. Covariant multenions will usually be distinguished by a " cross " (which, because of its position, is scarcely likely to be confused with a sign of multiplication), thus

a x , %x , x,

read " alpha cross," etc.

(2) Covariant and comixt linities will have no distinguishing mark. Contravariant and contramixt linities will usually be distinguished by a cross.

(3) Contravariant and covariant linities will be denoted by Greek letters ; contramixt and comixt linities by Roman capitals.

(4) a, 7ft, , ax, /3x, 7 x, will generally denote " dummy " vectors. (Edding uses " dummy " in an analogous sense for a positive integer.) Dummies are always (below, though not of necessity) treated as constants in that they are not affected by the differentiations of v«

The duty of a dummy contravariant or covariant multenion may be generally described as " to occupy, and thereby indicate, a situation into which an actual, previously defined, contravariant or covariant multenion, may legitimately be inserted."

§ 12. The Invariant Forms of any Multenion Expression.-77has been defined as a fundamental vector linity. It is also used for the corresponding extensive linity, a multenion lin ity ; and similarly for 77*. Since in (2), of § 11, ds?= (rpdf,if dp in our manifold takes the place of the dp of multenions in a Galilean manifold. [In a Euclidean manifold all the dimensions are real. In a Galilean manifold some are imaginary.] More generally, at a given position of the manifold, rpqwhere where q* is a covariant multenion, takes the place of q in a Galilean manifold. 77% and y~%qxwill be spoken of as the neutral forms of q and q*. Indeed the three q (contravariant), yq(covariant) and f q (neutra as only different mathematical representations of the same intrinsic property, of the manifold ; just as in an ordinary dynamical system, the whole system of velocities or instead the whole system of momenta equally will serve to specify the instantaneous motion.

Multenions and Differential Invariants.

Corresponding to any multenion expression such as Ya . (rs) which has a Galilean interpretation there are always the three forms r, s now being contra variant) in our manifold (contra., co., neut.).

V-*Va ■ [vW^bivWs)],V^a • [ ], and Va . [ ].

Tlie first and second of these three can alvjays be expressed in such a shape that only rj and not ??* occurs explicitly ; though the expression may be very complicated as in the example chosen. It is these final forms from which

■rp has been explicity banished, that are regarded as invariantive.

To show this, first note as an example, that the product pqr can be first modified by expressing qr as a sum of terms such as . Va^Yjr and then p . qr can be similarly modified. AVe have then merely to find the invariant forms (two, namely, covariant and contra variant) of Y [The reader is reminded that in our notation v, w, have homogeneities «, b, c, respectively, or have the forms

u = Y ap, v- Yw- Y ar.]

Let a be greater or equal to b and let v be contra variant. Then most frequently all we require to note are the two cases = 0,

V~^a+bViuViv = Y a+bUV,= Y a+bVuV'v>

V~i Y a-b7jiur)h - Ya-bUyV,- Y

and here of course we may have u - x or = r)~1v x in each of the forms.

To deal with the general value of c in we use Now we know that if ti, t2, ... t,, are n independent contravariant vectors at a point, then their normal reciprocals ?i, f 2, ... ?Mare n independent covariant vectors. Also we know that if (a, /3x) is any expression-bilinear in two vect

(£, 0= - 2 (To, Ta).

a= 1

Hence if we insert £, ^ into two places in an invariant form, of which one is occupied by a contravariant vector and the other by a covariant vector, the lesulting expression will be invariantive (meaning, " of invariant form").

Now it is easy to see by reduction to primitive units that

Va+l - 2cUV= [(-)4c(c+D/c !] Y a+b- 2c. Y a- cu{J')

where as usual ^ stands for and £ (c) for Y CQC\

Hence the contravariant form of Y a+b-2Cuv is

[(_) ic (C+i)yc jj y a+6_ 2c ^ y y (2)

fiom which the covariant form may be written down, and either u* = rju or v = 7]V or both may be used in place of u and v. In my MS. work I

am in the habit of using f°r K %cc\ Q£c^« Thus (2) would be
(c !)-1Va+j-2c • Va-cU&c)Vb-cVtk (c)v.Aspecial case of (2) often required the contravariant form of Y2gxb' where w and w' are contravariant and M Y2 is

V ,(Y ir«)(V ivW),(3)

or more generally, for infinitesimal rotations generally, the contravariant form of Va&m is

V A V ^ M V * -^). ' (4>

§ 13. Normal and Incident Components in Invariant Form.-These components occur very frequently in relativity. Thus the energy-momentum linity is density multiplied by an incident component, at any rate in its kinetic or primitive form. Normal and incident components serve to separate: (1) vector potential from electrostatic potential; (2) momentum from energy; (3) force from power; (4) current density from charge density; (5) magnetic induction from electrostatic force; (6) magnetic field strength from displacement. One should always be prepared to recognise these resolutions.

In quaternions if ais a given vector and c an arbitrary one, components with reference to a, namely the normal component

N qC = a-1Yac = aY . /a 2

and the incident component

Ioc = a-1Sac = aSac . /a 2.

As will be familiar to users of the quaternion method, it is these components each multiplied by a2, rather than the components themselves which we shall expect to meet with frequently in analytical expressions. The two

Nacr = Nor = aY ac. Iac = Ic a Sac,

may be referred to as the normal and incident expressions of c with reference to a. With regard to these four linities N0, Io, N, I we have as follows

No2 = N0, Io2 = I0, N0Io = 0 = IoN0
No' = No, V = To, N0+ 10 = 1 = No' + 10'
N2 = a2N, I 2 = a2I, NI = 0 = IN
N' = N, I ' = I, N + I = a2 = N' + r

Similarly in a Galilean manifold any multenion q has two components N0<?, I o qnormal and incident to a given vector a. Let

n
q= X iv,where w - Y ;

then when the Unities Nw, N0w, etc., have been defined, N<?, N have also been defined. We have

(Galilean IS Iw= aYc+i*w, I vj - ^

manifold.) = oTl Y c+\ctw,I

[If e is a vectorium of homogeneity a vector a has a normal component e_1V6+i€o- and an incident component e-1V6-ie<r with reference to e. W ith reference to e,w has many, not merely two, components given by

w = e-1V6+c€«?-!-e“ 1Vj+c-2€tt?+....

The terms on the right of this equation are all, separately, but not obviously, of homogeneity c.j

Next let a be a given contravariant vector at a given position p of our manifold and q an arbitrary contravariant multenion at the same position and let w - Y cq. Also let uv, qv> w v be the neutral forms and covariant forms of a, q, rwespectively, that is to say,

r/^ci = a,, = »ax 'j

7)1W = Wv = T)~%IVx V (4)

77% - qv - J

The invariant form of a2 is a ,,2 or

Yo arju= «,,2 = Y qcix7)~1ax.(5)

With our present meanings (3) is not invariant in form. There are four modes of choosing the invariant forms of N and I, N0 and I0 and generally of any multenion Unities, (1) covariant, (2) contravariant, (3) comixt, (4) contramixt. We will take number (3) of these for the basis of the notation in connection with N and I. A comixt Unity means one which acting on a covariant multenion produces a covariant multenion. For this type of Unity it is obvious that the invariant forms of (3) are given by

N Y cctYc+ir}aWx = Y cV~l axY c+1ccxWx1

I w Y crjotYc-icnWx = Vca x V c-ir)~l uxWx

(6)

N0wx NwX/ Y qU7)U = Nwx/Y 0a x'?7_1a x

low* Il(7x/Yo ctria. = lw x/ Y 0IXX

(2) comes with this notation to be modified in two ways only ; a2 is replaced by Y 0u r]cn

or Y qo.xTj~1ctxand in place of N' and we have a new kind of conjugate ^N't?-1, t)1't)~xinvolving the fundamental Unity rj. Thus now have

N2 Yburjci. N, I2 = .1, NI :

N, VI'V~X= I, N + I = VoV (N' + I')Y .,}■ <7>

[Note that if we put v^'v' 1 =
familiar properties

(<f> + yjr)c = <t>c+ 'fro(4>c)c =]

The types of linity we have to recognise are given in the following short list.

Linity Types.

Name	Co-variant.	Contra variant.	Co-mixt.	Contra-mixt.
Symbol 1	5	s*	X	Xx
			p x=	
Meaning...............		X		
		X		
	p * =	II	*	p
				=

Thus f converts a q(contravariant) into a (covariant) and simila the other types.

The two mixed Unities convert a multenion of a given type to one of the same type; the other two convert a given type to the other type. [In our present method the mixed Unities could scarcely be more unhappily named.]

Corresponding to any one of the types we have, of course, by means of rj, symbols of the other types with the same intrinsic significance. Thus, corresponding to the neutral form </> we have

£= 7]*<£*7*, %* = v *<f>v h

X = Xx = J

From the list and from (8) the following statements, in which £i, £2 mean two Unities of the type £, etc., are easily established :-

(1) fi|2 has no invariant significance, nor has £ixf 2*.

(2) X1X2 is of type X and X ixX2x is of type X x.

(3) Although by (1), for integral values of c, %c has in general no invariant significance; in particular when c--1 it has significance, f -1 is of type P , and p " 1 is of type f. X "1 is of type X, and X x" x is of type X x.

(4) A rational function/(X) is of type X ; and / (X x) is of type X x ; the corresponding neutral form in each case being /(<£).

(5) p is of type £, p ' of type p ; so that a similar remark applies to the self-conjugate and skew parts of f and £x . p and £x' correspond to <f>'.

(6) X' is of type X x, X x' of type X, both corresponding to </>'. Again ^ X V 1 is of type X and of type X x, both corresponding to <f>'.

(7) The identity satisfied by £ or by £x has no significance. The

nions

identity satisfied by <f> is the same as the identity satisfied by each of the following

x = fcr* = vV, xx= =

and all their conjugates and ^-conjugates.

(8) In the neutral form a finite rotation is effected by a vector linity (and its extensive) <f> for which = -1* This is equivalen V= 1 .

From the above it will be seen that if in a Galilean manifold we would deal with </>', <^>'+ </>, etc., then probably in the general manifold it is best to deal with £ and | x ; if in the Galilean we would use <£a,/(<£) etc., then in the general manifold the use of X, X x is indicated.

We need not write down the twenty-four alternative forms of the eight equations (6), but we may note how in terms of X, I the other types appear.

X, I are of type X ? j
X' = ?7-1X?7, I' = ri~l lr] are of type X x |
X?7, ly are of type £
?7-1X, ?7-1I are of type £x J

After the fundamental linity y,of type £, has presente three may be regarded as arising from

r \ * i$ r\ f r 1**

§ 14. Riemanns and Weyl'sManifolds.-Suppose + are for manifold the position vectors of three infinitesimally near points, P, A, B. We assume that for such small regions the metrics are intrinsically the same as in a Galilean system, and in particular that there is a definite shift of PA along PB, and of PB along PA, each called a translation, by which if A in the first shift comes to C, B also in the second shift will come to C. With arbitrary co-ordinates we cannot of course affirm that C will have the posi tion vector p + oi + f3,b ut instead that it will have

p + a + ZS-M/yS

where da%is to be regarded, not as a symbol of operation, but merely as the abbreviation of the words

^«T = " increment due to translation along a." (1)

From these preliminaries it is clear that

* ^«T/3 = d f* = -Ea& (2)

where Ea is a vector linity which is a function of the position vector p, and is also a function of *, linear in the latter; also Ea/3 is symmetrical in « and /?.

The word translation implies something about constancy or otherwise in interval or in " the square measure " dsof eq view we are to assume that in general a translation is such that it is impos sible for the square measure to remain unaltered when by successive steps a closed path has been traced. In such a closed path beginning and ending at a given point, there will appear a change in the square measure depending on the point and the path.

The square measure then is assumed to change according to the equation

daTV0 fivfi= - . V0Xa, (3)

where a and /3 are arbitrary and with the same meanings as before, and X is a vector, a given function of p.Thus a, /3 must be covariant. En/3 is a vector neither covariant nor contra variant, which need not surprise us, because more than one position of the manifold has been used in its description.

It is clear that 77 is arbitrary as to a scalar factor and X correspondingly arbitrary. More definitely, nothing is altered in the meanings given if (1) 77 is multiplied by a positive scalar h, any function of position, and (2) X is altered to X+ v log h, as appears from equation (3).

Important qualification of the last sentence.-77 here means the original vector linity denoted by 77. Whenever we speak of altering gauge by multi plying ?; by ht his restriction is to be understood. It is clear that yvj thereby changes, not to hiqw,but to hcr)W.

" Into all expressions or formulae, which exhibit analytically true metrical properties of the manifold, 77 and X must enter in such a manner that invariance subsists (1) when there is arbitrary transformation of co-ordinates (p replaced by p) ; and (2) when 77, X are replaced by hr), X + v log A, what ever positive scalar function of p, h may be."*

Since (3) is true, and also what (3) becomes both when /3 is changed to 7 and when changed to /3 + 7, we get the corresponding bilinear form

djV ofivy - -Vo/3'77 . V0Xa, (4)

where u,/3, 7 are all arbitrary. In the translation denoted by , 77 suffers the increment -VoaV-7? and and 7 the increments -Ea/3, -Eay. Hence

-v 0a y 9 . V0/3??9y-VoEa/9?77-Y0/37;E0y = -Vo/3777 . V0Xa.

Put 77E. = r a, (5>

so that Ta/3,like Ea/3, is symmetrical in a, /3. Thus

- Vo a{3 + ay)= V0a(v9-X) . Vo/37797.

* W eyl, ' Raimi, Zeit, Materie,' 3rd ed., p. 110. English translation, 1922, of 4th ed-(Brose), last sentence of p. 123.

ons

Write down the two equations obtained by cyclically changing a, /3, y, change the signs of the second and third, and then add the three equations. We obtain

2VoaIVy = Voa[(V9" V)Vo/3?797""Vo/3(V9-X) . Vo7(V9"-^) *

or

ryy = £ [-Vo/3(V9-X) . 1797 -Vo7(V9- • 17»£ + (V9"-*>)Vo£wy]*(6)

For Riemann's manifold we have merely to put X = 0. Since frequent reference must be made to this case of Riemann's manifold, we will put for the values of Ea and Ta when X = 0, Ea and Ta respectively.

From (6)

(r , + r /) y = -v 0£ (v » -*-) • wy> CD

(T ^ F f) y = V i[V a(V9-tyvvP • 7]* (8)

Thus T fy is obtained from (6) by reversing the signs of the second and third terms on the right. It further follows that T /y regarded as a Unity of /3, instead of y, is self-conjugate.

This, and its converse, can be shown to be true of any such function YJ3 which is symmetrical in a and /3.Thus we may put for the i Y,,/3, icY o'X'ty'cftwhere yfrc is a self-conjugate vector linity, or

Ya/3 = XicVoPfc* V^ ■

whence Ya'/3 = X ^ caV0i/3 C= 1 ^

showing that Y a'fi is self-conjugate with respect to a. Each involves £ n (n-f-1) scalars, so that Y a/3 involves \ n2(n + l) scalars; when = 4 this number is 40.

When Ya/3 is not symmetrical (9) still holds, now being not self conjugate, and the number of scalars being nz.

We shall from this point onwards take for granted, as is not difficult to prove, that at a given point it is always possible so to choose our co-ordinates that all the \ri*(n-f 1) differential coefficients (\79, yd) we zero ; and independently to choose the gauge (h) so that X is zero ; therefore also to make Ea zero for all values of a. [Brose calls h the calibration ratio. Selection of a definite h he calls calibration.]

§ 15. Absolute Differentiation.-Let r be a contravariant vector given over an assigned path or over a region of n or fewer dimensions. If the region is of fewer than n dimensions, we may still assign arbitrary values over a region of nd imensions containing the given region, so that we can continue to apply V to it.

In passing from the point p to the point p + x, where a is infinitesimal, the ordinary increment -Y0«V- T will denoted by d * \, and the excess of this above d j r will be called the absolute increment and denoted by dj^r. Thus we have

daAr = da°T-d j r = + E aT~|

= -V ,, « v . t + E. t (1)

We are about to show, as might be expected, that this absolute increment, daAr, is contra variant. If

X= -v o( )V • p

and the change from p to p causes the multenion function of position to change to q', and the multenion linity to change to <£'; then the nece and sufficient conditions that q should be (1) contravariant, (2) covariant, are

(1) 2' = X2> (2) 2' =

The necessary and sufficient conditions that <jf> should be (1) covariant, (2) contravariant, (3) comixt, (4) contramixt, are that should be equal to

(1) X~'4>X~1' (2) X<t>X>(3) x ,_W >

All these may be put in infinitesimal forms which are generally easier to apply. Let

p - p + 8p = p+e, %' = 1 + = 1 + ^o,

(q = q + 8q, (f> = (f>+ 8(f).

Then the above six conditions (necessary and sufficient) for the several types of invariance become

(1) 8q= Xoq, (2) =

and in the case of <£, 8(f) is equal to

(1) -% 0 ^ -^ X o , (2) %0^ + ^Xo', (3) -%</<£ + <£%</, (4)

and in order to make the tests we have

%o = - V 0( ) v • = -V 9 V 0( )e9. (3)

[% is applied to general multenions by the rules applicable to extensives; by those applicable to subextensives.]

First apply the test to EaT ; is SEaT = %oEaT ? Instead of working this directly it will be found easier first to find SVoaiyy from the equation pre ceding (6) of § 14, in which observe that, since V0a v = V0a'v'> 3 has no effect on V0«v> V0/3v, V07V- Further for a, /3, 7,

3(" > 7) = %o («, 0,

and for 77 8y = -Xo'v-VX<>-

It is thus easy to prove that

SV oar^ = -V0/3y9 . \ 07Vs .

and thence that

8Ep7 = %oE/sy-V0/3v • Voyv • e> (4>

so that Ear is not contra variant. Next we have

8 . da°T-Xoda°T = -S(V0aV • T) + %o(V0aV • T)

= -Y0^v • %ot + %o(V0«V . r)

because 8 does not affect Yo«V> and - X^T- Thus

8 . da°T = %0<4°T-f Vo»V9 . V0TV9 • <?9

whence finally

8daAr = %o^aAT>

or daAr is contravariant.

In one respect our taking of equation (1) to define has been misleading. If we follow on from it and equation (4) § 14 in a natural manner to define daxo x where crx is a covariant vector, the meaning arrived at will be on the whole inconvenient. For the moment we may take it as a sufficient reason, for this statement that if equation (4) § 14 is to hold for any scalar invariant function of position, x,then d jx being different from zero, if is equal to da°x - djx , absolute and ordinary increments of invariants must differ. We reject the suggestion of equation (1), then, except for the incre ment of a contravariant vector. It is convenient to introduce a new increment, the invariantive increment, ddgr.For t we will take

daT- d j r - -EaT,

but to fix the meaning for other cases we will take that

d^x = 0 (6)

when x is an invariant scalar function of position. This gives

0 = djVoCT^T = Vo (daVo<txE0r.

Since t is arbitrary we get

d 1**= E#, crx.

Finally in all cases we put

daA = dcP-da1 = -V0«V • -da1. (8)

The strongest reason for this meaning of daK remains to be given. It is the only increment of the kind which is truly invariantive for truly contra variant and covariant vectors ; vectors which besides their usual relations to change of co-ordinates are also invariant for change of gauge. If r is thus truly contravariant tjt cannot be truly covariant, and therefore we ought not to base our conception of the standard increment of <rx on supposing it possible to put <rx = rjr.

Nevertheless the following up of the meaning of da a x from equation (4) of § 14 has its uses, and we shall make one important application of it when later we consider curvature. It is not difficult to show that

daTa*= Ea crx -<rxV0«A,, ?? = - -Y0aV . (9)

(so that if we had continued to take da

A= da

as in the Eiemann manifold, that da\y= 0)

above daAyis not zero, as in a Kiemann manifold, but instead

dg^l7= -V quX .

when rj is used in its primitive sense as a vector linity. When used as an extensive linity in pw we get

daxp .w -

We will now find the absolute increments of the two kinds of multenions and the four of multenion linities. Mixed Unities interpreted as subextensive, have invariant meanings, but not the other two types. Thus if <£ is contramixt

(j>\^x/3y - Yz(f)a8iy +

is of invariant form, because then <£a, y are of one type, and is not of such form when is covariant, because then ft, y are not all of one type.

Let Ea and Ea' have the subextensive meanings as well as their primitive vector linity meanings. Thus

E.w = -VcEa£Vc_i£w, E«'w = -VcE«/fVe_i %w. (11)

We at once have

d^p - da°-

p dalp

d^q*= d a°q * -d^q*= -

Eor the four linities f x, X, X x, treat of the first as an example of all four. The meaning of e£aA|? is given by

dax^ . p - adA

Here %p is a q*so that

daA(Zp) = -Yo«V • (&)-E*%p.

Hence d p tj. p = -Y0a v 9 . Ea'£p-£E*p.

Thus

d, = - Y 0aV . £ - E a'£ - £ E a

■dSt* = - V 0av • P + E a£x + f xEa'

(14)

<4AX = - Y 0aV • X - E / X + XE.'

d.AX x = - Y 0*v • X x + E aX x - X xE«

It is instructive to re-write in the present notation the four kinds ot B(f>
above

= -%o'£-£xo |
*r = » r+ r » ' i (1B>
SX = - x ,'X + X Xo' |
sx* = » x * - x * x „ j

Doubtless the complete parallelism between (14) and (15) has some simple

explanation. Of the four results in each set, the second follows from the
first by putting £x = £_1 and the fourth from the third by putting
X* = X'.

We obtain six very general formulae showing how to convert such
expressions as (\7y, ?9X) or (v 9>p9) to invariant form [provided («*, or
(a*, p) are already bilinear and invariantive] by one common process.
Introduce a second notation for daA, thus

(la A q = <u, d= <f>a, (

Put £ in place of a in (12), (13), and (14). Thus

i t p d = (V9, p d + iE (12a)
it to*)= (V». it / (13a)
tox)E~
i t &) = (V9, td + it -1
i t h x) = (v 9, h x) + i t E i f + r E j ') !
(14a)
itX{) = (V9, X9) + (fc - E / X + X E^) j "
i t V) = (V9, X9x) + (^, E jX x -X XE^) J

(12), (13), (14) are particular cases of (12a), (13a), (14a).

As an example of the use of these equations let us suppose we have with
Galilean co-ordinates established the equation T9y 9 = 0 where T is the
(complete) energy momentum linity ; a linity which in the general manifold
is best taken as cornixt; a self-conjugate vector linity (self-conjugacy means
for a comixt linity T = ?;T 'v~l>that is »;_1T or T77 is self-conjugat
ordinary sense). The (truly) invariant form of T9y 9 is given as Tf£ by the
third of (14a). For generality we will at first not assume T to be self-
conjugate. We have

T«£ = T9V 9 -E /T £ + T E ^ .

To evaluate E^T£, write down Ea'/3X = from (6) of § 14, change
a, to £ T£, and simplify. Thus

*VT£ = l v, (^ - iT - T V 1) (V 9 -X)-1 (V9-X) (16)

When 7j_1 T is self-conjugate we get

E*'T£ = -M V 9 -X)Y Or ^ - 1Tr. (17)

Further, since . d\<f)\= -Vof<£ ld<f>£, putting T = 1 we get
Ef'C= Jc-'vJc-^nX.(18)

In equation (16), the skew part of rj~lT alone occurs in the first term, and the self-conjugate part alone in the second. t7~'T is contravariant. Let us put = £* and use the second of (14a) for the further reduction

&xr = & xv 9 + E fr £ + r E / £

We may use (18) to modify E /f. E ^ |x^ happens to be particularly simple

as it involves only the self-conjugate part of | x ; let £* = f 0x + £ix where £0* is the self-conjugate part and | i x the shew part. We find that

E ^ x£ = (17a)

and f£x£ = &'1(&f XMV9-2 n \) + r)~'+i^o^»?9|:

Linities of the form (f>+ §V0£'</>£' frequently present themselves in relativity, (f> being mixed and generally self-conjugate in the mixed sense.

Note that f ^ a n d X^x^ have no invariant meaning because £/3x and X X/3X have none. We thus do not seek generalisations of £979 and X9XV9- V is a symbolic covariant vector because Vo dp\? = Vo

(17a) is found again below and numbered (22), in the course of illustrating a generally useful process.

Returning to the generalisation required of T9y 9 = 0, when T is comixt and r)~l T is self-conjugate, we get

(^ T)9 (v 9 - ^ a) + K V 9 -^)V 0r W 1(^ T) r = 0. (19)

It is clear that JcT, a vector linity density as it is called, rather than T itself, should be regarded as the fundamental function. (19) is our standard form on the hypothesis that T is truly invariantive. If it were of power a in the sense about to be explained, the first V9 in (19) should be replaced by V9 - d\.

If an invariantive symbol has such a meaning that on change of gauge (hy for 77, A. + for X) the symbol is multiplied by ha then it is said to be of power a. 77 is of power 1 when .used in its primitive sense as a vector linity; in yvj, 77 is of power c; powers that we find when v applies to 77 (power 1) as in Ea and T*, v appears only in the combination - \ \ when v applies to h (power \ri) we always find it in the combination v - Thus in (18), although in a Eiemann manifold we should find E / f = Jc~\Tc, we must expect in the Weyl manifold to find instead

3V(f= &-1 (v -

The reason for this is easily seen. Thus if q is of power a, (y 9, q^) is of no

definite power at all, but as is quite easy to verify (y 9-a \, q9) is of power a.
To be lynvariantive
trui a symbol must be of power zero.
An important case of this kind occurs in connection with the earliest invariant expressions we met, namely,

Vi+ivy*.

Our standard forms for these are based on the supposition that u and v* are each of power zero so that V i+iV ^ has not to be modified; but in the

other case kuis of power \n and our new invariant form should be A:-1Ya- i (y - %n\){ku).Now (12a) and (13a) show us that our standard forms in these two cases should be

Vfl_i yn + Va-i£EfW, V6+i vtf* -Vj+iS'E/'y*.

These must be the same as those we have just been considering so that we must have

Va-ifE = /fc-1Va-i [(v - . u\, (20)

V*+1^ V = 0. (21)

These identities can be established directly, but not very easily.
The specially simple form of r a+ IY given in (7) of § 14 furnishes a second useful result from (16) for it gives

(i y + r ^ T c = w - 1t (V9- x)

where i y ^ -1 = E$'. Thus from (16)

i y r]T r = - i ^ (i ? "1T + T/iri)(v ® -X) + i(v 9 -X)Y o riw " 1T?. (22)

First we have the specially simple case T = 1, namely

= V9V~1V9-k~l'Vk + ±(n -2)\. (23)

Another very simple particular case is when 7?-1T is skew. The right of (22) is then zero, and ti~xTis of the form Vi®( ) where is a contra »V2. Thus

iy Y iw ^ = 0.

Since Ta = ??Ea we also have E^Yico^ = 0. This last may be regarded as a special case of (21) and suggests two other forms of (20), (21). Operate on (20) by Y0w* ( ) taking c - a -1; and on (21) by Yo) taking = 1.
We thus get

E /Y c+1w * £ = k~xV c+1w(v -

E^VC_ iio£- 0, (25)

(18) is a particular case of (24).

§ 16. Curvature.-If (a, /S) is any function of two vectors, linear in each then

(*> £)-(£ , a) = (£, YxCVaa/3 . $]) (1)

as is seen by putting (a, ft) = - (£ £V0«£) and (ft a) = -(£ , aY<ftf). Thus when

(a, f t = - (f t a) then (a, f t = | (£ Vx [V2a/3 . £]). (2)

The application of these two equations should be constantly borne in mind in what follows, even when attention is not called to it.

Similarly for the various invariant forms of infinitesimal rotations of a vector,

Vi coot*,Vi ft)x«, Vi a*.

Let us calculate the increase in y,a contravarian starting from the point p,along the four vector sides parallelogram in the order «, f t -a, -f t a and /3 being contravariant. Afterwards let us calculate the similar increments of 777. We shall put V2«/3 = co,and the increments due to translation round the parallelogram wi be denoted by d jy , .djy Since the difference of two contravariant at a point p is itself contravariant it follows that is contravariant, and similarly djy7is covariant; and it is obvious that they must both be linear in both co and 7. We shall, therefore, put

dJrjy = djy7 = = fl^y, (3)

Cw being a contramixt vector linity and the corresponding covariant linity, which may be called the curvature Unities. Along the side + a the increment in 7 is -Ea7. Along the side -a, 7 has changed to 7 - and d -J - -d j has received the increment -VoySy. ( ). Thus, these two side contribute increments

-Ett7 + (Ea-Vo/3v • E,,)(7-E ^),

= -Y0/3y • E,,7 -EaE|37.

The contributions of the other two sides are obtained by interchange of a and and reversal of the sign. Hence, noting that Ea is linear in a,

C<o7 = d j y - -Eg . v)wrS)7 -b(-EaEp-f E^E,,) 7.

[Since d j = da° -d j, and since da°taken round t zero we get that Co/r = toj3- rpawhich shows that absolute di different directions are not commutative.]

In treating of r/y in place of 7, instead of -Ea we have to write E / -VoA« for d j. Noting that if x,y are two scalars (-YOA«, -Y

- (E.' +)xE

(/+ y)+

we see at once how (4) has to be modified. Tims

7= d jy y = E9. ylU)V./ yf- (-E

where co* - V2y \ (6)

[co and wx are of very different natures. is a contra variant is a given truly covariant function of position ; in relativity <ox is magnetic induction cum electrostatic intensity, A being vector potential cum electro static potential.]

From (4) on putting <w= V2*$, and noting that EaY is symmetrical in « and 7, the following “ cyclic rule ” is seen to be true for each of the two parts of Cw7 separately

Cyaa/*7 "bCv2/3ya + bV2ya/3 = 0. (7)

Remembering that ft* = yC *, inspection of (4) and (5) shows

f t w+ ft< / = - Vo&>wx .

or CU)+ i]~1CJ t) = -Yow&)x J

The equation in + is even simpler than that in r a + IY ; and for a Biemann manifold ft* -1- ft*’ = 0.

The first of (8) shows that ft* = -|Yo&)ft>x.?? + Vift>’( ) where &>’ is a covariant linear function of <w. Hence

ft*Y = -|Y 0ft)ft)x . ?;y + Vin7ft)Y, (9)

where nr is a covariant „Y2 Unity. The first term on the right of (9) gives the ratio of stretch -|Y 0o)ft)x ; the second term the rotation vsco; suffered by 7 by d j.

To find the properties of the Unity cr first transform (5) to the following,

ft</y = r 9. Viw(v9- a)’ 7 + (IY tt1r 3- r / ^_lr a) 7 - 777. v 0cocox. (io)

I will indicate in outline the mode of doing this. The first term on the right of (5) is

TyYo/^v?- ^7Y0a v 9

and the second term is

(- + Tf},vi~1Y a’) y = (IYi7_ ir p-l y ^ -1 !^) Y+ two terms

of which the first is -IY??-1 (Tp + i y) 7 and the second is -f JY) y.

It will now be found that the parts depending on y in (r,s + r y) and (Fa+ IY) cancel with the following parts of the first term.

IY 0 r 1) ^ • V0£V9-i y . Y0«y9.

Collecting the parts not cancelled we get (10).

In the first term and the third on the right of (10) there are terms of the form (y 9, A9). These give

hvy • Yo&>&)x - 2 Y i • ^ 2 (^ Y ift)y 9 . A9) 7

= -2^7 • Y0ft)ft>x -4 Yi . V2(7?Yift)V9 • . \?9)y

-i Y i . Y2(77Yift)V9 • X9- . y 9) Y-

The last of these three parts furnishes the skew part mi, of nr, given by

ciift) = iV 2(V^ft)x . rjYi^oi). (11)

The neutral form of this is the suggestive expression |V 2&>X&>. The remainder of nr, denoted by roo> the self-conjugate part, is

wow = -£V2[(vd-x) i?9Vio>(v ®-^)] + i 'v a to r{ V l r j '.'Vi®s,-y
- i V 2(i;Yi&)V« • Yt+^Yi&jXg. y 9) J

So far we have divided £lwy into three parts, such that the corresponding parts of Cwy are truly invariant, (1) the part self-conjugate in 7, namely,

_ | 7?(yY0&)(Ux ; (2) the part Yjnxia>7 in which mx is skew and (apart from the fundamental Unity y)only involves a* as with the first part; (3) the V1TOOW7 where m0 is self-conjugate. Part (3) obeys the cyclic rule ; (1) and (2) together, but not separately, obey the same. The cyclic rule for ot0 may be put, as will be shown later, in the form

v 4. = 0,

so that CTOinstead of having the full number

| . \n { n -1) . 1) + 1]

of scalars appropriate to a self-conjugate ,,V2 Unity, has a smaller number because of the n (n l-

)(w- 2){n- 3)/4 ! scalar equations of just written. The number of scalars of is thus). For the parts (1) and (2) the §?i(?i-1) scalars of &)x are required. The total number of scalars involved in Cwis thus

-fa n (n-1)

When n -4 the number is 26.

Putting Vi£w = Vi&ox = the two parts (1) and (2) together may be written as

+ i v 1v 2(rMx^r«o) 7

(13)

= i (- y. VoSXx - . V0^ox7+^xV 0^ 7

It is now desirable to separate mointo three p covariant as to change of co-ordinates though not as to change of gauge, namely, the parts of degrees zero, one, two in X.

ma, of course, gives precisely the Riemann terms, obtained by putting X = 0 in any expression for m. Thus from (12)

maw = -| y 2 . y ^9Y3<wv9+-2 Y j& ry1? l r foYi&)^. (14)

Next write down the terms involving differentiations of X and make them invariantive by (13a) of § 15, using, however, E^' in place of E / as we desire invariance only for change of co-ordinates, and we desire not to introduce quadratic terms in X. Thus we obtain for the probable value of mb

mhQ} = -| Y2 (^Yawyc, . X9-f ??Yi &)X9 . y 9)

(15)

+ iY2 . E^X + ^YjoE^X . 0

Before verifying that this contains all the terms linear in X, not differentiated, find the quadratic terms, Put A,, for the terms of IY which contain X, that is, Aa = r a-Fa. Thus

gtcw = -AY2X7?V 1&)X+ i Y2£oA / »?" 1AfOVi® £ (16)

To simplify this put w -Y2«/3 and work in the neutral form, finally re to invariant form. The result comes out that 4cyc&> is the normal expression of to with reference to X (in neutral form XVsXeo). Hence

nreco = 4rjV2XY3^~1A.co = 4V2^- xXY 3X77®, (17)

ciTjj, nr*, vscare all separately self-conjugate, and all separately obey the cyclic rule.

We will now show that (15) must contain all the terms of (12) which are linear in X. Inspection of (12) shows that these terms form a function (Ye> Vo,Y «) which is linear in all four constituents. If there is any term of this nature which has not been included in our three parts vsa, ctj, gic, such term must be covariant as to change of co-ordinates. Now (vo, X, <o) can only be made covariant in this sense by the addition [first of (14a) of §15] to it of

(£ -Es'rj-r£jE, X, a>) - -(£ T y + 1$,X, co) = Y &>)•

This addition reduces the supposed absent term to zero. Hence there is no such term.

We now proceed to the two successive " contractions " of ftw or Cw. The first contraction yields a vector linity denoted by ft or C in its covariant or contramixt form. The second contraction yields a scalar, an invariant of course, F.

^y^pV"1^is a covariant vector because the subscript £ occupies the place of a contra,variant vector and the second £ occupies the place of a covariant vector. Put then

W = C/3 = r,-iQ/3 = C v a v -'Z (18)

F = - Y o ^ - 1^ = - Y O£C£. (19)

^Note, that of the four symbols ftw, Cw, ft, C, the only two are truly invariantive are Cwand ft ; ftwis of power 1 and C of powTer -1; F also is of power -1.

Let us now obtain the terms contributed to ft/3 and to F by our five parts oi ftw, that is, by the first term in (9) and the four parts of gt, namely, gti, via,vsi, r,c. lo r the latter four other than &ra work details in the neutral form, and at the last step put the result into invariant form by the rules of §12. Where first differential coefficients of X occur, attend in the detailed working only to the terms in which the differential coefficients occur, and in the last step use equation (13a), of § 15, modified by reading E / instead of E / to obtain the invariant form.

Let us denote the parts contributed by the first term of equation (9), to 12 and F by 122and F2, and those contributed by rsh ct m, wc, by 12i, ftBl 12&, 12c and by Fi, F,, Fb,Fc, respectively. Thus we find

122/3 = - £ V aa>x/3, fli/8 = i (» - 2) Y i » ^

r• (^50)

(122+ 12i)/3 = \ (n- 4) Vi&>x/3 . J

These are the only parts of 0/3 that are not self-conjugate,and they are purely skew. It is aremarkable fact that 0 is self-conjugate for the case 4, and for no other value of n. Of course, this accidental circumstance, ifwe are to call it such, isfortunate for the theory of relativity. From (20), it at once follows that

Fa = Fi = 0. (21)

It is easy to show that when a term of cr is self-conjugate the corres ponding part of 12x of 12 is self-conjugate. &a, vsb, vjC) are self-conjugate. We find

4 tlb/3 = -2rj/3 . k~1YoVUcV~~l -(n -2) (VsAroA9.&+ AgVoV^ + 2E^,X)

F6 = - (n - 1) 'Yo

k~ V

In 12c the normal expression of /3 with respect to A occurs thus :

FIJ3 - i (n -2)Y ir)-1\V 2\yfi^

Fc = i (u - l) (n - 2) Y 0A ^-1AJ '

In a Biemann manifold rna completely specifies n w, and therefore H, 12wa, and Flare given by any expressions for 12^ and 11 by putting A = 0. The most useful form of 12a is derived from (5). To get 12/3 from (5) w'e have to put a = £, ??y = f. Thus we obtain

12/3 = E ^ 'v s -E /E /C + E /E /r+ V o /S v . E # + V x / 3 ® * . (24).

Here note that the first three terms are just those, when we put E / = T, that were under discussion in equations (16), (17), (18) of § 15, and (24) may be modified accordingly. Our immediate purpose, however, is to put A, and therefore <ox zero, so as to obtain Fla. Noting that E /£ = by (18) of § 15, we have

H,,/3 = E ^ v » - E /E /C + (E / + V0/3v •) Y log k . (25)

This is the form (26'3) given in Eddington's Report on Relativity, that is to say, Eddington's Gb,c is our Y^ibFlaic [see equation (7), § 17 below]. (25) gives

F« = - V oiT ^ V vb-E&/E*/ro + [E*/ + Yofv •]) (26)

Returning to equation (24) let us find a useful expression for It is convenient on occasion to use the following notations

= r,«, ,- > r ,, - > = r,«,

but this is a little cumbrous for our present purpose. Let us put
' =
where yaj3x and vp* [see (24c) below] are put for the self-conjugate and skew parts of and avoid the double suffix notation ^ 0X, £/ji *. Putting = 0 in (24) .we have
c a/3= • v i°g
[Put - r (= i v - (P i + y) = p ? + v 0rv • v-]

xVe + E^ V £ + /ir 1V
= + *T1(Vo/3V . v log y
whence, using (17a) of § 15 for which involves but not vp*,
C a/3= h~x (^ x)9V9 + '?-1 (W 0 X+ i V0^ 9 ^ x 0 V9 + V~l (V o fiv • V log fc).
(246)
dnd vp*being |iy" 1(±^/s + lV) '?" 1 are written down from (7) and (8) of § 14. Thus
V 7 X = |Vo^V3 • ('T 1)s7x 1 ,
V 7 X = - i V i [v -'V iV m fi • 7 X] ^ *
From (246) we may write down lc¥a in the form -7cY0^Ca^ and we shall reach an expression useful for reducing, later, 8 (kFa) resulting from changing 1] to 7]-f 8rj.
In what immediately follows let us work with the neutral forms, only occasionally stopping to point out the slightly more complicated invariant forms. In the case of a vector Unity, </>, w'e may say that there is a definite part, namely, -n~lV0£cf)Z,which gives rise to the contraction -V0£<££; the second part, <fi + ?t-1V0£</>£, has zero contraction and involves only n2-1 scalars as against the full number, n2, for a vector Unity. Similarly, it is possible to separate out from any vector expression 7) linear in each constituent a definite part (n-l) -1 (V2£Vi <»7,£) absorbing scalars from the full number, \ n z (n-1) required to s remainder of (00,7), namely
<®, 7) - (» " i r 1(Va?V1®7, 0
has zero contraction. The part thus furnishing the contraction C of Cwis
(n - l)" i C = (» - 1) '1CViQjy
and it obviously satisfies the cyclic rule (and would do so whether C</y itself did so or not).
The following form of the cyclic rule for Cw7 may be noticed:-
< W v C = 0. (27)
This is true because by (1) of § 7
VtfVe»7 = V2/3y V0«£+ A Y /a V ^ F V8«/3 Vu7?

(27) is closely connected with a generalisation of (1) and (2) above. The cyclic rule is an identity which is combinatorial in a, 7 ; that is to say, the sign of the expression equated to zero alters by interchange of any two of the three «, ft,y. If («i a2,..., ««) is any expre combinatorial in the vectors, it can always be expressed as a linear function of aa(a\ It is actually equal to

(a I)"1(gi, & .... 5.-1, Vi«a(a) (28)

(28) may be proved by mathematical induction by proving the general connecting link in the following successive equalities, a being, for example, taken as equal to 4,

(a, & 7, 3) = (2 I)-1(£ Vi . V2«/3 . £ 7, 3)
= (3 I)"1(6, & Vi • V8«£7 Va(i k 3)
= (4!)-1 (£i, £3, Yi . V4 «/373Y3^1^^j).

This is readily done when it is noticed that

Y iST 'V 4«/373 = 2±«V *$8'"£78

and the like (&"' = V,&6&).

Since V2^Vi &)7 -V2 (V;j&o)7 = -ceiVo£y

we have Cw7 = CVio>7 -Cy.cv^y? (29)

whether or not Cw7 satisfies the cyclic rule. If the rule is satisfied we liave further

Cw7 = CVa(v1fo,)75 (88)

This may be obtained from (29) or directly from the cyclic rule in its original form th u s:

C w7 = C y 2aj37 = C v 2a y /3 - C v« = C'V2(V1^) y ^

by (1) above.

It is useful to modify the cyclic rule for a self-conjugate namely, mi,, or mc. Thus, using ma as an example, we have = 0, or operating by Y03 ( ), where 3 is a fourth vector,

VaajSroaVayS + Vo • V2a7

The expression on the left is combinatorial in «, /3, 7, 3. Taking a, j3, 3 to be any four of the primitive vectors i\, ... say, ih 13, ii, we have succes sively the following three statements :-

2 [ti"1*2~1'3~14~12Vo(tii2) nrs (t3t4)] = 0,
• Yo(V,rx6)ara(V,&{;) = 0,
V4(VaCi&)w«(V8$i?») = 0, (31)

as mentioned above between equations (12) and (13).

From equation (27) onwards we have been at no pains to put our results in invariant form. Nevertheless (27), (30), and (31) are already in invaiiaiu- form, and (29) yields

Co/y = CViowpy- (29a)

§ 17. Comparison withTheory of Tensors.-Let i >•••?«) an)Tcon variant or covariant function linear in each of the multenions ... each one of which is also contravariant or covariant, not necessarily all of

one type. Thus, for instance, is such a covariant function of «,/3,7 x, two contravariant vectors, and one covariant vector. In the pages above we have defined various increments of q due to changing where « is an arbitrary infinitesimal contravariant vector

(dj, ,fda°, daK)q

and we might add, when q is of power the increment proper, also of power c, dax*q - daBq - (dax + c \ 0aX)q. The increments of $ of these are all formed on the same principle. Thus take daK as the type and use the abbreviated notation daxq - qa. Then the increment <£a of <f>is given by

</>a(?i, 72, ...

a

= [</>(7n ?2, ... ?a)]a- 2 qa, ... qc* ••• qa)

C = 1

where qca occupies the position of qc in <f>(qi,q2, ...)• Once having obtained our increment, we may cease to regard a as infinitesimal, so that rfaA becomes, not so much an increment as a rate of increase multiplied by the magnitude of a, though above it has always been called an increment.

All this is on the model of the Theory of Tensors. In that theory, how (qx, ,zq..., qa)recognised is the scalar form </>(“i><*2, ...,««) in several vectors, some of which are covariant and the rest contravariant. The present writer marvels at the great results obtained by expounders of the Theory of Tensors, since the basis is ’such a limited concept.

The precise translation from our presentation to that of the Theory of Tensors and viceversd is to a certain extent arbitrary, on account of vagueness, especially in the matter of sign, that writers on the Theory of Tensors seem to permit themselves. The following will be found to form one consistent system. Let Aa, B6 be two contravariant vectors, a and (3, and let Xa* be a covariant tensor of the second rank, with the same intrinsic meaning as £, a covariant vector linity. AaB* seems to he taken to mean the same as BjA^ in the Theory of Tensors, but I cannot see how this can be consistently done, . and in the system here given AaBt and B*Aa, when interpreted as vector Unities, must have different meanings.

With the basis as given in § 11 above, we have as follows. Let

X

R

£ x =* v/3,

II

(i>

| P = 7 ?- 1p - 1, X = p - 1, X * = v- ’ l (2>

n 0 0

V = - a2= 1 10 P dcca- ˜ V°taV’ (3)

a ° = v 0o r 1*, Aa = "Volac*’ > (4)

gal = Vo iaV

H

,gah = 9a = V0i«4˜\ (5)

\bc, n] = Vo0iEtl ic>t}f == Voia 1Etl<c (6)

= Vo latib, Xf = V0 iaX lbX«* =

The equation X ab = A„Bft consistently with the above and the related equations may be interpreted to mean as follows

X a = A.Bfo £= v*V0( ) ˆ

Xa* = A.B6, X = ,;«V0( ) 0 «*V0( ) 0. (9)

Xa6 = AaB?’, p = 0( )/3. (10)

Now (8) and (10) definitely tell us what BjAa and BL4a must mean instead of representing f and p respectively, they represent f ’ and p ’ and B6Aa is similarly naturally interpreted as ??/3Vo()a, that is as

Lastly we have

n

(f, ?) = - 2 («„, <„-■). (U >

a= 1

It is with these interpretations that equation (25), § 16, was stated to be equivalent to Eddington’s (26’3).

I may add here that Eddington’s equation (23) for Baˆ gives

B abch= -V0fc*“1C!y#,»uta

where Cwmeans Cwa,that is the part of C,* obtained by putting X =

It will be seen that I interpret the mixed tensor to mean a comixt, not contramixt, vector Unity.

18. Prof. Eddington’s recent ork*[Re-written Septembe

w

Suppose ABC is a triangle whose sides AB, BC, CA are geodesics. Let the point 0 of an infinitesimal lattice-work pass round the triangle, starting and ending at A, and in each stage of the journey let the lattice-work suffer " parallel displacement " only. When the circuit has been completed, then :-according to Euclid and Galileo the lattice-work will be found occupying its initial position exactly; according to Riemann it will be found turned through an angle in a plane containing 0 ; according to Weyl it will

* ' Roy. Soc. Proc.,' A, vol. 99, p. 104 (1921).

be turned and changed in size, but not in shape ; a
will be found turned, changed in size and also in s
meant no more than this, his manifold, as will be s
mathematically equivalent to Weyl's, but he does me
primary manifold there is no criterion for shape at
square from any other parallelogram, or to distingu
parallelepiped. In Eiemann's manifold the fundament

invariant quadratic form (squared interval), and "
mathematically derived idea. In Eddington's manifol
is that of parallel displacement, and an invariant
mathematically derived idea. When n = 4, Eiema
Eddington's foundations require ten, thirteen (nine
quadratic form and four coefficients of a linear fo
point to be given, respectively.

Eddington then starts with Ea/9(= E^a, - Ea/9 m
due to the parallel displacement
without any fundamental
our previous work and resu
continue to he valid in Eddin
case where our old reasoning ceases to hold in the
is equation (4), § 15, which was proved by aid of
still true because the condition that t, a contrava
parallel displacement /3 is

-V o ,#v • T = -EjsT,

and this is an intrinsic condition indepe
Infinitesimally change the coordinates as
S(Vo/9v) = 0]. Thus

-Vo/9V • (Xor) = -S (E^t) ;

and here the left transforms to

~%o(V0/Sy . t) -V0/3y9 . ^ 09t = -^ 0E^r-V0/Sv9 • YoaT

so that

8(Epr) = XoE(3t + V 0/3V9 • %09T.

Ihis is equation (4) §15 with r in place of <y. We may then make the very important use of that equation which we made before, namely, if r is any contravariant vector,

daAr = -V0«v • T+ EaT

is contravariant and may therefore be appropriately called the absolute increment of r. [Prof. Eddington does not show in his paper that absolute differentiation thus holds good in the " in- " sense in his manifold.]

Our chief disability from absence of a fundamental t) seems to be that there is now found no place for the ideas associated with normal and incident components (and expressions) and with rotations. Y a+buv and have their invariantive equivalents, but apparently in general has not. We can still speak of a and /3X being normal when Y0a/3X = 0 and this leaves us with (what we met with in § 10 before ?? was introduced) the ideas of normal and incident components with reference not to a single r but with reference to tc, some one of n given independent contravariant vectors

n, t 2, .r n. In a word the normal reciprocals of n, t2, ... r ncontinue to have invariantive meanings.

For present purposes the contraction we selected in § 16 to pass from <JUto Clwas unhappily chosen as it depended on ij. We will here, therefor change the contraction to another. The previous definition of fl/3 was that it was obtained by setting a, y equal to in ; our new meaning of Cl'a. is obtained by setting /3, y x equal to £ in Cy^'yY Thus

Previous Cl(3 =

Present Cl'a = C y^' £

In Weyl's manifold the change in meaning is slight. The old and new self-conjugate parts are the same, but whereas the previous skew part of Cl/3 was -4)V1(yx/3, the present is -\nY\a>*(3. This simplicity of con nections between old and new meanings may be proved from the identity [equation (8), § 16].

■tyCy^y + Cyi0Lp'vy- -Vo«yxYr2«/3 . i]y,

by setting a,y equal to The identity is the one which expresses that our lattice-work, after its journey round the circuit ABC, is changed in size but not in shape.

Our equation for Cw (4) of § 16 still holds while rj is unassigned, and from it

a / y x = - E 9.vlWV/ y x + (- E / E / + Ea'E.8) y x. (2)

Putting here Yn»v = «Vo/3v~ /3V0av and using (1) we get

Cl ol = E9.a' V9+ Vo«V • E ^ - E ^ E / f + E / E ^

The first, third and fourth terms on the right are all self-conjugate because E a/3 = E^a. Hence the skew part of Clu is - f YiaYavE/t.

By its original meaning Cwy is the difference of two truly contravariant, or as Eddington says in-contra variant (would not prefer-contra, variant, be better ?) vectors at the same point. Hence Cw, Cl and Y2v E ^^ all have the " in- " property and -|(H + Cl')is fitte mental of a Riemann manifold. Prof. Eddington, therefore, identifies k V Orf/>(n + n')dp with Einstein's quadratic form, the squared in terv al; c being an absolute constant. I do not propose to follow him in this respect, but to show instead, that as soon as it is recognized that " lengths " can be compared (even in Weyl's limited sense that equality of lengths can be postulated ata point only), then Eddington's theory is mathematically equivalent to Weyl's ; and instead of merely one simple quadratic form there are several available to identify with Einstein's.

It is now necessary to distinguish between Weyl's and Eddington's

meanings of Ea, Ow, G and related symbols. For Eddington's meanings we shall use E a, ECW, Hfi. For Weyl's we shall use Ea, WCW , WG.

The following three modifications of notations occurring in §§14 to 16 above are desirable ; (1) The letter G will be used in place of the letter F, because F is wanted below for another purpose; (2) the " contraction " to pass from Cwto O (both of power zero), will be altered in meaning as already described ; (3) the symbols 2, 1, 0, a, b,c, used as suffixe (former F), will be shifted from the right-hand bottom corner of the letter affected to the left-hand top corner. The following scheme will probably be helpful:-

EC<u

, j---------------- 1----------- 1----------- 1

*Cw :tC(o WCut

(144) (36) ,---------1-------- ,-------------- i

2Cu. 'Cut "Cot

1------------- ¥------------ > (20)

(6) |---------L------- 1------

«C» *Cut

[In each of these four lines there is one symbol equivalent to a tensor of Eddington's, or to a tensor of Weyl's. ®CWis equivalent to Eddington's *Babch, wCtt to Weyl's F abch,°CW to Weyl's *Fa6cA, and "C*, to both Edd IW 1and Weyl's EnjcA.] In the scheme the E of ECWimplies " in Eddington's sense" ; the W of WC,, implies " in Weyl's sense" ; further, the scheme implies the following three defining equations:-

ECW= 4Cft)+ 3a)+ wCU

wCa> = 2Cw+ 1C(U+ °C(0,

°C<o = aCw+ 6C<o + cCw.

We may write x for any one of the ten symbols E, W, 4, 3, 2, 1, 0, a, b, c.. lo r seven out of the ten meanings of *0^,, namely, for all except "C^, 6CM, cC<0, it is true that xQui is of zero power as meant by Weyl. The three aCw, ^C^,, cCw, are invariantive for change of co-ordinates but not for change of gauge. The numbers in brackets (144), etc., imply that B0u, involves, when 7i = 4, 206(= 144+36 + 6-f 20), independent scalars, that WCW involves 2(t(= 6 + 20), independent scalars, and soon. [It is explained below that the statement that 3CWinvolves 36 scalars, merely means that 3CWis fully known when the 36 scalars of Y aft below are k known when the 144 first derivatives of the 36 scalars are known. In a purely algebraic sense 4CWcannot involve more than 96 scalars.]

In what now follows Eddington's symbols

1' ah', * I 'aftA * H aft, I ' aft; 1 aft, E a ft, c, ^aft, c, 2

are the equivalents of our

Ea, ECW , m , H Eft + E* n (E^ Vfft Ap7, Ef'G

respectively.

In our interpretations, Eddington's " increment due to parallel displace ment " has to be distinguished from our (for Weyl's manifold), " increment due to translation." We use f?aPD and d j for these phrases. As with (5) of § 14 above, we have

d ^Yvfir]7 = -Y0»V9 • YVfygy -V0E a^ y -Vo/S^EaY.

Prof. Eddington remarks that as this is the difference of two invariants it must be invariant, and therefore must be of the form 2Yoa'Pj3y (a more general form than ours, after Weyl, --Yo/fyy. Y0Xa in equation (4) of §14), where ^ y is a covariant vector linear and symmetrical in ft and y (contra-variant vector dummies), given by

24yy = -VaYo/S^y -E/?/y -E /?;^. (4)

^ 7 can easily be expressed uniquely as the sum of two parts, a con tractile part involving n scalars and a non-contractile part involving %(n + 2)n(n-1) scalars. The contractile part is - and is so named because it is the sole part of "'kyy which is responsible for the contraction a vector. Thus

2 % y = -X V o/V / + 2T>y (5)

where X is a covariant vector (namely, and T^y is a covariant vector, linear and symmetrical in ft and y, and satisfying the vector equation

T ^ = 0 . (6)

[Digression.-If Zaft si contravariant (and similar remar covariant), and does not satisfy the equation of symmetry Za/3 = Zp*, there are three kinds of contraction (derived from the scalar form Y0y xZa/3). The corresponding contractile parts of Zaft are

-y-1£i • Yo&jZ^ • Y0a?7 ft,r)~lZi'%. Y0 -

the three contractions themselves being

- • Yo^a Z(i&>%

sV~l£

The first and second of these contractions are equal when ZJ3 - Zpx.]

The X appearing in equation (5) is the X we met with in Weyl's theory. If we examine what change occurs in 'Pgy, or better in due to change of gauge, we at once find that in ?;-lxPgy the whole change falls on the X part, no change at all occurring in y~lTpy ; that is ^ _1T 37 is of zero power.

Calculating from (4) the value of

Vo«iVy = Voa'JVy- VoP'Sfya

we at once find that

A37 = (^ - ^ V) - ' ! a)

and E37 = E37 + ^37 (8)

where Y37 = y - l i T , y - T I y - r y'/3) (9)

and E37 is exactly as in § 14. On the right of (7) the two bracketed terms furnish the skew part of A3 and the third term the self-conjugate part. It will be easily seen that

Y/ £ = - T ^ - ^ = 0 . (10)

From (9) Y37 is of zero power. It follows that it is just as legitimate to name -E37 the increment of 7 due to the translation /3 (d^y) as it is to name -E37 the increment of 7 due to the parallel displacement fi(dpvv,y). In other words Eddington's manifold is simply Weyl's in which there is supposed to exist a given Y37 of zero power and satisfying the equation Y j'f = 0.

With our present notation [see (4) of § 16]

KCw7 = duPX)y - -E9.vJa>v9Y"i'(-E aE3 + E 3E a)7. (11)

Putting E a = Ea + Ya we get a part involving Ea, E3 only. This is, of course, our former

VVC(U7 = d^y = -E9,vlWV97-f (-EaEg-f EgEa) 7. (12)

Next we get a second part

3Cu7 = (- Y aY3 + Y3Ya)7 (13)

and a third part which may be written

4C » 7 = -(Y a 7)* + (Y37)a (14)

where ( )p and ( )a denote absolute increments in the sense developed in § 15 above for Weyl's manifold. (14) follows from the beginning of § 17 from which is deducible

(Ya7)3 = - Y 9a7 . Vo/3V9 + E3Ya7- YaE37- Y ^ * .

With n = 4 (13) involves the thirty-six scalars of Ya/3 and (14) involves their 144 first derivatives (y). The full meaning of our scheme for BC(U, given above, will now be evident.

There are numerous simple quadratic forms available to identify, in Pro!. Eddington s manner, with the square of Einstein's interval. Not only may we contract (12) and select its self-conjugate part °fl, but we may similarly contract each of (13) and (14) in two different ways and this by no means exhausts our available simple forms. Besides (13) and (14) we may similarly use (YaYp + YpYa)y and (Yay)p+ (Ywhe contraction is available (put a., @- £

Returning to the use we made of (3) above, we see that the skew part of rQa is -^V iaV avE ^- Since Y?'£ = 0,

E = v logfc - (15)

by (18) of § 15 above. Hence Prof. Eddington's identification of Y2v E / ^ i n the electromagnetic field is virtually the same as Weyl's identification of V2y \. Also note that EX1-wf2 (= 3H + 414) is self-conjugate.

One consequence of Eddington's suggestions is that we may quite likely desire to identify Einstein's squared interval with something very different from our fundamental form Yodprjdp.Let Einstein's Y 0dp9dp. We may wish to put 6= | (EH + Efl' or 6 -i (30 + 3H), the last making 6 depend solely on Ya/3. With any such identification we require symbols dependent on 6, as are WH, etc., on ij. We shall, therefore, henceforth use

9, l, F j3, ©£, Cw, O, ay, G-,

in the 9 manifold (Riemann manifold) in the same way as above V, E k, * Tp, we., have been used in the rj manifold (Weyl manifold). The relations between 9, l, eF p,tc. are precisely the same as those between v,h aE|3, ai>, ac w, an, am, ®G, ("E* aTphere replacing our earlier Ee, Tp

If 9 is identified in some such manner as just suggested the determinant |0 | may become zero. |# | = 0 and |0 -1 | = 0 represent loci. Do the loci if really existent, constitute the boundaries of electrons and atomic nuclei ? On crossing the boundary do we pass to a region with four real dimensions or to a region of two real and two imaginary dimensions (two-dimensional time and two-dimensional space) ? Is it on the other hand meaningless to speak about crossing the boundary, much as to an inhabitant of hyperbolic space it would be meaningless to speak of passing across his space's boundary, the absolute ?